Course notes

Area and antiderivative

For f continuous and positive on $[a, b]$, $a \leq b$, the integral $\int_a^b f(x)\ \mathrm{d}x$ is the area under the curve, above of the axis, between $x = a$ and $x = b$, expressed in d ·air units.

Fundamental Theorem

If f is continuous on an interval I and $a \in I$, then $F_a(x) = \int_a^x f(t)\ \mathrm{d}t$ is the antiderivative of f who is zero at a. For every antiderivative F,

$$\int_a^b f(x)\ \mathrm{d}x = F(b) - F(a).$$

Every function continuous on an interval has antiderivatives. An integral is a number; an antiderivative is a function.

Proof in the positive increasing case. For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After division by h and taking the limit, the rate of change tends to $f(x)$. For $h < 0$, monotonicity gives $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. By integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

Dividing by $-h > 0$, we get

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

Both bounds tend to $f(x)$. By continuity, $F_a' = f$. Any other antiderivative is $F_a + C$, hence the calculation formula at the bounds.

Properties

For continuous functions and bounds in the same interval:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For a function of any sign, the integral is defined using antiderivatives. The geometric area between the curve and the axis is $\int_a^b |f|$, obtained by splitting at the sign changes. Between two curves, it equals $\int_a^b |f - g|$.

Example. $\int_{-1}^{1} x \, \mathrm{d}x = 0$, but the area between the line and the axis on this interval is 1.

Mean value and integration by parts

For $a < b$, the mean value is $\mu = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x$. The rectangle of height μ and width $b - a$ has the same algebraic area.

Integration by parts

If u, v are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

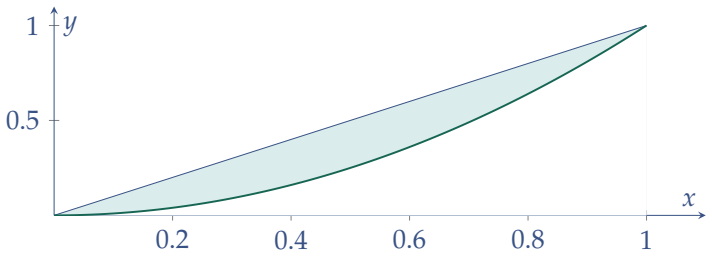
Proof. Integrate $(uv)' = u'v + uv'$ then isolate the required integral. The bracket is indispensable.

Example. With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e-1) = 1$.

Rectangles and sequences of integrals

For f continuous and increasing on $[a, b]$, with $a < b$ and $n \geq 1$ and $h = \frac{b-a}{n}$, the sums $h \sum_{k=0}^{n-1} f(a+kh)$ and $h \sum_{k=1}^{n} f(a+kh)$ bound the integral. Their difference is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, start with the sign and comparisons of the integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \frac{M}{n+1}$; the the squeeze theorem gives $I_n \to 0$. Integration by parts can also give a relation between I_n and I_{n+1}.

Convention for index zero. In $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, the expression x^0 denotes the constant function 1, y included at the bound 0. Thus $I_0 = \int_0^1 f(x)\,\mathrm{d}x$; no indeterminate form needs to be resolved.



The area between $y = x$ and $y = x^2$ on $[0,1]$ value $\frac{1}{6}$.

Linking area, antiderivative and approximation

8.1 Two areas, one integral

First use: S59 and S61.

Worked example · A change of sign

On $[0;3]$, $f(x) = x - 1$ is negative before 1, positive after. The integral equals $[x^2/2 - x]_0^3 = 3/2$. The geometric area is $-\int_0^1 f + \int_1^3 f = 1/2 + 2 = 5/2$. The integral involves cancellation between two signed contributions.

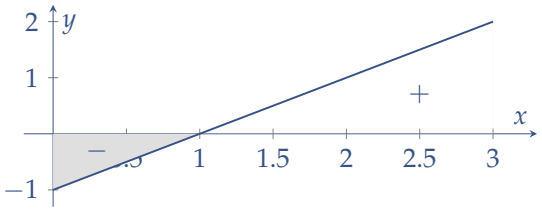


Figure 8.1. For a geometric area, the two regions add up; for the integral, their signs are involved.

A1 · Immediate application For $g(x) = 2x - 2$ on $[0;3]$, calculate the integral and the area between the curve and the axis. How does this relate to the values in the example?

8.2 Why l ·aire gives a antiderivative

First use: S61.

Property · Fundamental theorem

For f continuous on an interval I and $a \in I$, $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is an antiderivative of f. So, for any antiderivative F, $\int_a^b f = F(b) - F(a)$. The general case is accepted without proof; the positive increasing case is proved below.

Proof. Suppose f is continuous, positive and increasing. For h>0, the comparison with two rectangles gives $f(x) \le [A(x+h) - A(x)]/h \le f(x+h)$. For h<0, it gives $f(x+h) \le [A(x+h) - A(x)]/h \le f(x)$. By continuity and bounding, the difference quotient tends to f(x); thus $A' = f$. Any other antiderivative F differs from A by a constant; since A(a)=0, $A(b) = F(b) - F(a)$.□

Worked example · Recovering a derivative

For $A(x) = \int_0^x 2t\,\mathrm{d}t$, the area is x^2 when x is positive. The antiderivative formula also gives $A(x) = x^2$ for negative x, with the orientation of the integral. In both cases $A'(x) = 2x$.

A2 · Partial guidance On $[0;+\infty[$, set $A(x) = \int_0^x e^{-t}\,\mathrm{d}t$. Give A(x), A'(x), the variation and the limit. Interpret the limit as an area.

8.3 An equivalent height, with its units

First use: S63.

Definition · Mean value

For $a < b$, the mean value of f on $[a;b]$ is $m = \frac{1}{b-a}\int_a^b f$. If f is positive, the rectangle of width b-a and height m has the same area as the region under the curve.

Worked example · Flow rate and volume

A flow rate $d(t) = 2 + t$ L/min for three minutes provides $\int_0^3 (2+t)\,\mathrm{d}t = 10{,}5$ L. Its flow rate mean is 10.5/3 $= 3.5$ L/min. The volume and the mean flow rate do not have the same unit.

A3 · Immediate application Calculate the mean of x^2 on $[0;1]$. Draw the equivalent rectangle. Explain why this mean value is not the mean of the two endpoint values.

8.4 Draw and check the bounds

First use: S63 and S66.

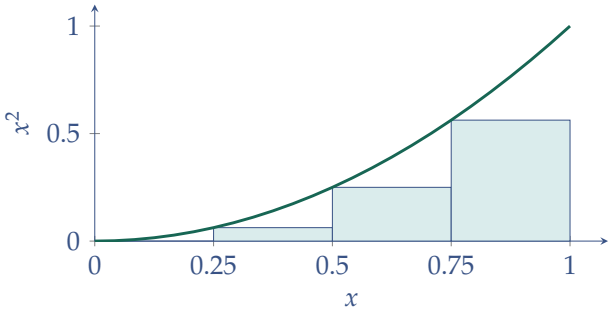


Figure 8.2. Left-hand rectangles. Complete the drawing with the right-hand rectangles and the line at height 1/3.

Worked example · Two intervals

For x^2 on [0;1], two rectangles to the left give $L_2 = \frac{1}{2}(0^2 + (1/2)^2) = 1/8$; those on the right give $U_2 = \frac{1}{2}((1/2)^2 + 1^2) = 5/8$. The integral 1/3 does lie between these bounds.

A4 · Investigate For $f(x) = x^2$ on $[0;1]$, divide into four equal intervals. Calculate the sums on the left and on the right; justify their positions relative to the integral. Then show that, for n intervals, their gap is 1/n.

A first step: In the difference of the two sums, the interior values cancel.

8.5 Choosing what to differentiate

First use: S66.

Property · Integration by parts

For u, v of class $\mathcal{C}^1$ on $[a;b]$, $\int_a^b u'v = [uv]_a^b - \int_a^b uv'$. The formula results from the integration of $(uv)' = u'v + uv'$.

Worked example · Two successive integrations by parts

For $I = \int_0^1 x^2 e^x \,\mathrm{d}x$, differentiating a polynomial lowers its degree. First $I = [x^2 e^x]_0^1 - 2\int_0^1 xe^x \,\mathrm{d}x$. Then $\int_0^1 xe^x \,\mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \,\mathrm{d}x = e - (e-1) = 1$. Hence I=e-2. Write the bounds at each step avoids forgetting a constant.

A5 · Choice of strategy Calculate $\int_1^e \ln x \,\mathrm{d}x$. Justify your choice in the integration by parts and check the result by differentiating an antiderivative.

8.6 Bound before looking for a formula

First use: S66.

Worked example · Bounds without an explicit antiderivative

Let $J_n = \int_0^1 x^n/(1+x)\,\mathrm{d}x$. Since $1/2 \le 1/(1+x) \le 1$, $1/[2(n+1)] \le J_n \le 1/(n+1)$. Both bounds tend to 0, so $J_n \to 0$, without calculating Jn for each n.

A6 · Synthesis For $n \in \mathbb{N}$, $I_n = \int_0^1 x^n e^{-x} \,\mathrm{d}x$. Show that (I_n) is positive and decreasing. Establish $e^{-1}/(n+1) \le I_n \le 1/(n+1)$, then its limit. Does integration by parts give a relation between I_n and I_{n+1} ?

A7 · Refute Give a non-zero continuous function on $[-1;1]$ whose integral is zero. Which additional hypothesis, together with continuity, would allow us to conclude that a function whose integral zero is identically zero?

One minute, spoken aloud. Explain on the figure why "zero integral" and "zero area" do not mean not the same thing.

Delayed review · S79. Calculate an area between trigonometric curves while keeping the search of the intersections and the sign.

Investigation

Bounding an integral with rectangles - 35 min.

We want to bound $I = \int_0^1 x^2 \,\mathrm{d}x$. For an integer $n \ge 1$, we split $[0;1]$ in n intervals of the same length. Let L_n the sum of the areas of rectangles whose heights are the values of x^2 to the left endpoints, and U_n the one obtained at the right-hand endpoints.

1. For $n = 4$, draw both families of rectangles, then calculate L_4 and U_4.

2. Justify that $L_4 \le I \le U_4$.

3. Write a Python function with argument n which returns L_n and U_n. Determine algebraically $U_n - L_n$.
4. Determine the smallest integer n for which this bracketing has a width at most 0.01.
5. Calculate the exact value of I using an antiderivative and check that it lies within the bounds obtained for $n = 4$.